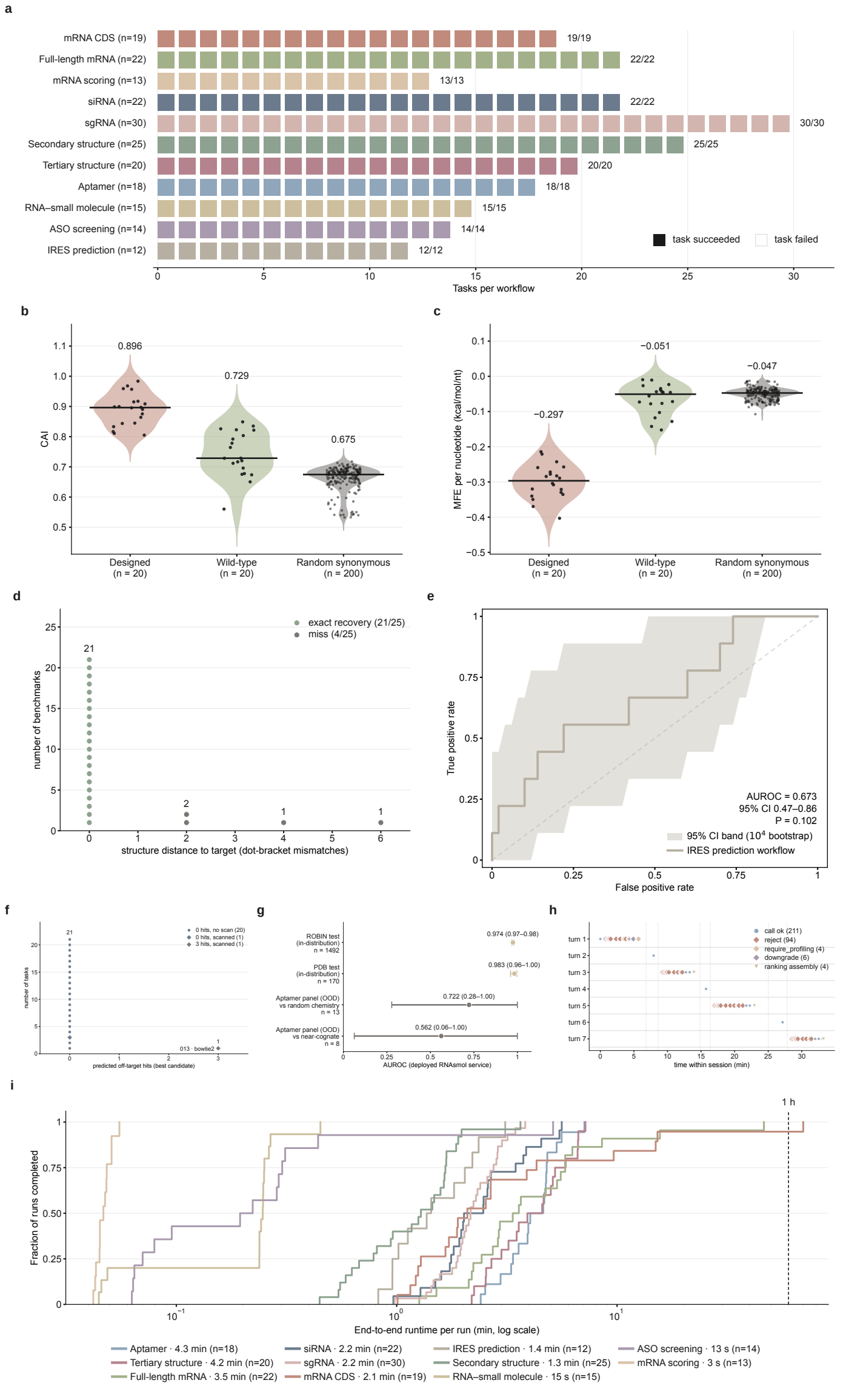

a
mRNA CDS (n=19)
Full-length mRNA (n=22)
mRNA scoring (n=13)
siRNA (n=22)
sgRNA (n=30)
Secondary structure (n=25)
Tertiary structure (n=20)
Aptamer (n=18)
RNA–small molecule (n=15)
ASO screening (n=14)
IRES prediction (n=12)
19/19
22/22
13/13
22/22
30/30
25/25
20/20
18/18
15/15
14/14
12/12
task succeeded
task failed
Tasks per workflow
b
CAI
0.896
0.729
0.675
Designed (n = 20)
Wild-type (n = 20)
Random synonymous (n = 200)
c
MFE per nucleotide (kcal/mol/nt)
−0.297
−0.051
−0.047
Designed (n = 20)
Wild-type (n = 20)
Random synonymous (n = 200)
d
number of benchmarks
exact recovery (21/25)
miss (4/25)
structure distance to target (dot-bracket mismatches)
e
True positive rate
False positive rate
AUROC = 0.673
95% CI 0.47–0.86
P = 0.102
95% CI band (10^4 bootstrap)
IRES prediction workflow
f
number of tasks
0 hits, no scan (20)
0 hits, scanned (1)
3 hits, scanned (1)
013 · bowtie2
predicted off-target hits (best candidate)
g
ROBIN test (in-distribution) n = 1492
PDB test (in-distribution) n = 170
Aptamer panel (OOD) vs random chemistry n = 13
Aptamer panel (OOD) vs near-cognate n = 8
0.974 (0.97–0.98)
0.983 (0.96–1.00)
0.722 (0.28–1.00)
0.562 (0.06–1.00)
AUROC (deployed RNAsmol service)
h
turn 1
turn 2
turn 3
turn 4
turn 5
turn 6
turn 7
call ok (211)
reject (94)
require_profiling (4)
downgrade (6)
ranking assembly (4)
time within session (min)
i
1 h
Fraction of runs completed
End-to-end runtime per run (min, log scale)
Aptamer · 4.3 min (n=18)
Tertiary structure · 4.2 min (n=20)
Full-length mRNA · 3.5 min (n=22)
siRNA · 2.2 min (n=22)
sgRNA · 2.2 min (n=30)
mRNA CDS · 2.1 min (n=19)
IRES prediction · 1.4 min (n=12)
Secondary structure · 1.3 min (n=25)
RNA–small molecule · 15 s (n=15)
ASO screening · 13 s (n=14)
mRNA scoring · 3 s (n=13)